

Linux Commands Cheat Sheet for SOC Analyst & Network Analyst

A beginner-friendly Linux command reference for log analysis, troubleshooting, networking, and cybersecurity practice.

1. Navigation Commands

Command	Meaning	Example
<code>pwd</code>	Show current directory	<code>pwd</code>
<code>ls</code>	List files/folders	<code>ls</code>
<code>ls -l</code>	Detailed list	<code>ls -l</code>
<code>ls -a</code>	Show hidden files	<code>ls -a</code>
<code>cd folder</code>	Enter folder	<code>cd logs</code>
<code>cd ..</code>	Go back one folder	<code>cd ..</code>
<code>cd ~</code>	Go to home directory	<code>cd ~</code>
<code>clear</code>	Clear terminal	<code>clear</code>

2. File & Folder Commands

Command	Meaning	Example
<code>touch file.txt</code>	Create file	<code>touch notes.txt</code>
<code>mkdir folder</code>	Create folder	<code>mkdir logs</code>
<code>rm file.txt</code>	Delete file	<code>rm test.txt</code>
<code>rm -r folder</code>	Delete folder recursively	<code>rm -r logs</code>
<code>cp file1 file2</code>	Copy file	<code>cp a.txt b.txt</code>
<code>mv old new</code>	Rename/move file	<code>mv old.txt new.txt</code>
<code>cat file.txt</code>	Show file content	<code>cat auth.log</code>
<code>less file.txt</code>	Open large file	<code>less auth.log</code>

Command	Meaning	Example
<code>head file.txt</code>	First 10 lines	<code>head auth.log</code>
<code>tail file.txt</code>	Last 10 lines	<code>tail auth.log</code>
<code>tail -f file.txt</code>	Live log monitoring	<code>tail -f auth.log</code>

3. Search & Filter Commands

Command	Meaning	Example
<code>grep word file</code>	Search text	<code>grep error auth.log</code>
<code>grep -i</code>	Ignore case	<code>grep -i failed auth.log</code>
<code>grep -r</code>	Recursive search	<code>grep -r error logs/</code>
<code>find</code>	Find files	<code>find . -name "*.log"</code>
<code>locate</code>	Quickly locate files	<code>locate auth.log</code>
<code>wc -l</code>	Count lines	<code>wc -l auth.log</code>
<code>sort</code>	Sort lines	<code>sort users.txt</code>
<code>uniq</code>	Remove duplicates	<code>uniq users.txt</code>
<code>cut</code>	Extract columns	<code>cut -d ':' -f1 file.txt</code>
<code>awk</code>	Pattern processing	<code>awk '{print \$1}' file.txt</code>
<code>sed</code>	Replace/edit text	<code>sed 's/error/warning/' file.txt</code>

4. Important Operators

Operator	Meaning	Example
<code>;</code>	Run sequentially	<code>cmd1 ; cmd2</code>
<code>&&</code>	Run if success	<code>mkdir test && cd test</code>
<code> </code>	Run if failure	<code>cd test echo failed</code>
<code> </code>	Pipe output	<code>cat file grep error</code>
<code>></code>	Overwrite output to file	<code>echo hi > a.txt</code>

Operator	Meaning	Example
>>	Append output	<code>echo hi >> a.txt</code>
2>	Save errors	<code>cmd 2> errors.txt</code>
&	Run in background	<code>sleep 10 &</code>

5. Permissions & Users

Command	Meaning	Example
<code>chmod</code>	Change permissions	<code>chmod 755 file.sh</code>
<code>chown</code>	Change ownership	<code>chown user:user file.txt</code>
<code>whoami</code>	Current user	<code>whoami</code>
<code>id</code>	User/group info	<code>id</code>
<code>sudo</code>	Run as administrator	<code>sudo apt update</code>
<code>passwd</code>	Change password	<code>passwd</code>

6. Networking Commands

Command	Meaning	Example
<code>ip a</code>	Show IP addresses	<code>ip a</code>
<code>ping</code>	Test connectivity	<code>ping google.com</code>
<code>netstat -tulnp</code>	Open ports/services	<code>netstat -tulnp</code>
<code>ss -tulnp</code>	Modern netstat	<code>ss -tulnp</code>
<code>traceroute</code>	Trace network path	<code>traceroute google.com</code>
<code>nslookup</code>	DNS lookup	<code>nslookup google.com</code>
<code>dig</code>	Advanced DNS lookup	<code>dig google.com</code>
<code>curl</code>	Send HTTP requests	<code>curl https://example.com</code>
<code>wget</code>	Download files	<code>wget https://example.com/file.txt</code>
<code>ssh user@ip</code>	Remote login	<code>ssh admin@192.168.1.10</code>

Command	Meaning	Example
<code>scp</code>	Secure file copy	<code>scp file.txt user@ip:/tmp</code>

7. Process Monitoring Commands

Command	Meaning	Example
<code>ps aux</code>	Show processes	<code>ps aux</code>
<code>top</code>	Live process monitoring	<code>top</code>
<code>htop</code>	Better process viewer	<code>htop</code>
<code>kill PID</code>	Stop process	<code>kill 1234</code>
<code>kill -9 PID</code>	Force stop	<code>kill -9 1234</code>
<code>jobs</code>	Show background jobs	<code>jobs</code>

8. Disk & System Commands

Command	Meaning	Example
<code>df -h</code>	Disk usage	<code>df -h</code>
<code>du -sh folder</code>	Folder size	<code>du -sh logs</code>
<code>free -h</code>	RAM usage	<code>free -h</code>
<code>uptime</code>	System uptime	<code>uptime</code>
<code>uname -a</code>	System info	<code>uname -a</code>
<code>history</code>	Command history	<code>history</code>

9. Log Analysis Commands (Very Important for SOC)

Command	Purpose
<code>tail -f auth.log</code>	Monitor logs live

Command	Purpose
<code>grep failed auth.log</code>	Search failed logins
<code>grep "185.220.101.5" auth.log</code>	Search IP address
<code>cat auth.log grep root</code>	Filter logs for root user
<code>grep -i error app.log</code>	Find errors
<code>awk '{print \$1}' access.log</code>	Extract first column
<code>sort file.txt uniq</code>	Remove duplicates

10. Compression Commands

Command	Meaning	Example
<code>tar -cvf</code>	Create tar archive	<code>tar -cvf logs.tar logs/</code>
<code>tar -xvf</code>	Extract tar archive	<code>tar -xvf logs.tar</code>
<code>gzip file</code>	Compress file	<code>gzip auth.log</code>
<code>gunzip file.gz</code>	Decompress file	<code>gunzip auth.log.gz</code>
<code>zip</code>	Create zip	<code>zip logs.zip auth.log</code>
<code>unzip</code>	Extract zip	<code>unzip logs.zip</code>

11. Package Management (Ubuntu/Debian)

Command	Meaning	Example
<code>sudo apt update</code>	Update package list	<code>sudo apt update</code>
<code>sudo apt upgrade</code>	Upgrade packages	<code>sudo apt upgrade</code>
<code>sudo apt install</code>	Install package	<code>sudo apt install nmap</code>
<code>sudo apt remove</code>	Remove package	<code>sudo apt remove nmap</code>

12. Cybersecurity Tools Commands

Command	Meaning
<code>nmap target</code>	Scan ports
<code>whois domain.com</code>	Domain information
<code>tcpdump -i eth0</code>	Capture packets
<code>strings file</code>	Extract readable strings
<code>file sample.exe</code>	Identify file type
<code>sha256sum file</code>	Calculate file hash

Beginner Learning Priority

First Learn

1. `ls`
2. `cd`
3. `pwd`
4. `cat`
5. `grep`
6. `|`
7. `>` and `>>`

Then Learn

1. `find`
2. `awk`
3. `sed`
4. Networking commands
5. Process monitoring

Simple Memory Tricks

Symbol	Easy Meaning
<code> </code>	Pass output
<code>></code>	Replace file

Symbol	Easy Meaning
>>	Add to file
&&	If success
	If failure
cd ..	Go back
grep	Search text

Practice Exercises

Exercise 1

Search for an IP in a log:

```
cat auth.log | grep 185.220.101.5
```

Exercise 2

Save output to file:

```
grep failed auth.log > failed_logins.txt
```

Exercise 3

Monitor logs live:

```
tail -f auth.log
```

Recommended Focus for SOC Analyst

Most important commands:

- grep
- tail -f
- cat

- `less`
 - `awk`
 - `find`
 - `ss`
 - `ip a`
 - `ssh`
 - `history`
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Final Tip

Do not try to memorize everything at once. Use commands daily while practicing in Linux or TryHackMe. Repeated usage builds memory naturally.